Dynamic Facial Expression Feature Learning Based on Sparse RNN

Ruicong Zhi[1,2]; Ming Wan[1,2]
1. School of Computer and Communication Engineering, University of Science and Technology Beijing
2. Beijing Key Laboratory of Knowledge Engineering for Materials Science
Beijing, China
zhirc@ustb.edu.cn, wanm@xs.ustb.edu.cn

***Abstract*—At present, dynamic facial expression feature learning methods are mainly divided into traditional hand-crafted features and self-learning features. The process of feature extraction in the hand-crafted features method is cumbersome and the recognition accuracy mainly depends on the quality of the features. Although deep learning method overcomes this problem, it is computationally intensive and poorly interpretable. In order to reduce the complexity of LSTM for sequence processing, this paper introduced a sparse representation method for feature learning, and the network structure termed SLSTM had the theoretical basis of ISTA. The SLSTM were applied on several facial expression recognition tasks, such as pain detection on Biovid database, and facial micro-expression recognition on CASME II and SMIC. The experimental results were compared with the classic LSTM network and most of the popular applied methods. It showed that our proposed method showed superiority in the comprehensive results with low computational complexity.**

Keywords—facial expression recognition; feature learning, sparse coding; RNN-based

I. INTRODUCTION

With the rapid development of artificial intelligence, people are increasingly pursuing a more intelligent and user-friendly human-computer interaction experience. Emotion recognition has attracted increasing interests recently as the basis for achieving this. Facial expression (FE) is a very important channel for human to express emotions, and a large part of the information is transmitted by it in daily communication. Facial expressions can intuitively reflect people's conscious and psychological activities, and are the most rapid expression of human inner activity relative to speech and body language. Facial expression recognition (FER) has also been well applied in many fields [1-3].

Although various applications emerge in an endless stream, due to the high feature dimension of the video sequence and high computational complexity, facial expression recognition based on the video sequence has many limitations in practical applications. Sparse coding has demonstrated its advantages in extracting features from high-dimensional data in many areas, such as image super-resolution [4], background modeling [5], preference prediction [6], etc. The essence of sparse representation is to represent as much information as possible with as few resources as possible. This representation not only improves the speed of calculation, but also enables automatic feature selection, i.e. weight of useless features is set to 0.

In this paper, we tried to introduce sparse coding to popular applied deep learning method, i.e. LSTM (Long-Short Term Memory) based on the Iterative Hard-Thresholding Algorithm (ISTA) [25] for dynamic facial feature learning on facial expression sequences. This network structure termed SLSTM can perform automatic facial feature learning on the video sequence and generate corresponding sparse codes, and finally obtained recognition results through the classification matrix.

II. RELATED WORK

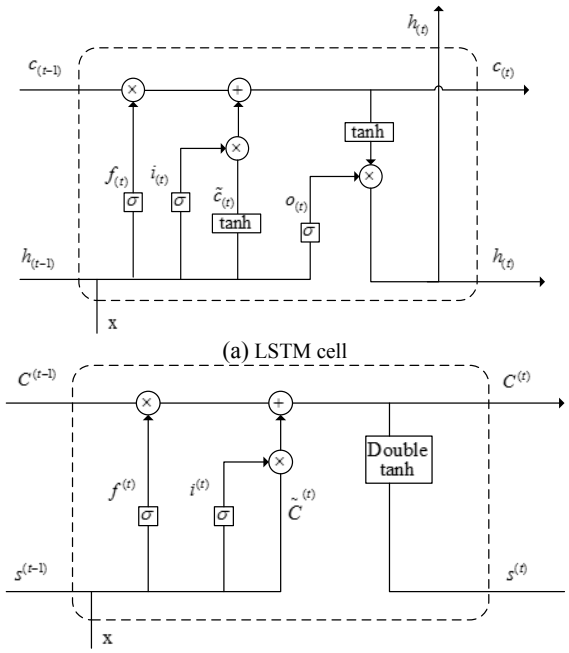
In recent years, there were many works on dynamic feature learning for facial expression recognition. One of the most traditional methods was to combine various hand-crafted facial features to pursue the improvement of recognition accuracy. Since deep learning had achieved remarkable results in the field of image recognition, more researchers tried to apply deep learning networks to facial expression recognition for automatic feature learning. Yu et al. [7] proposed a deep convolutional neural network (CNN) based facial expression recognition method, with multiple improved frameworks to further boost the performance. The network proposed in [9] consisted of two convolutional layers, max-pooling, and four inception layers as introduced by GoogLeNet. This proposed architecture got the outstanding performance on seven public databases. With Recurrent Neural Networks (RNN) being widely used in sequence tasks, a hybrid CNN-RNN architecture was proposed for dynamic facial expression analysis in [10], the CNN was used for frame feature extraction while the RNN was used to aggregate frame features. Similarly, the characteristics of facial expressions were encoded in two parts in [8]. As the first part, spatial image characteristics of the representative expression-state frames were learned via a CNN. In the second part, temporal characteristics of the spatial feature representation were learned with a LSTM. This network achieved outstanding results in the CASME II database. In order to further improve the recognition accuracy, increasing researches had begun to combine hand-crafted features with deep neural networks. Li et al. [11] proposed the 3D flow-based CNN model for video-based micro-expression recognition. The proposed network consisted of three data

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stream sub-networks that each sub-network had input data with one tube channel only: grayscale frame sequences, the vertical and horizontal optical flow. In [12], the network was based on two different models. The first deep network extracted temporal appearance features from image sequences, while the other deep network extracted temporal geometry features from temporal facial landmark points. These two models were combined using a new integration method.

For all of these methods that had been applied to facial expression recognition, the process of extracting hand-crafted features was cumbersome, and the feature descriptors were very susceptible to external conditions. Meanwhile, the method of automatic feature learning through deep neural networks, had not achieved the best performance due to its unexplained nature. In order to speed up the computational efficiency and explore the potential and interpretability of deep neural networks in the field of facial expression recognition, we tried to apply the sparse LSTM in this field.



(a) LSTM cell

(b) SLSTM cell [14].

Fig. 1. The unit of LSTM and SLSTM

III. LSTM WITH SPARSE CONSTRAINS

A. What is the SLSTM?

LSTM [13] has achieved impressive performances in dealing with image sequence for many tasks. Briefly speaking, LSTM is a variance of RNN, and the cell of LSTM introduces forget gates, input gates and output gates based on the cell of the basic RNN. The forget gate determines which information needs to be discarded from the cell state, that is, it can selectively retain information that it considers important. The input gate determines which new information needs to be entered into the cell state, and the output gate determines the output at this instant. Compared to RNN, LSTM can remember information over long periods and reduce the vanishing gradient problem. A basic LSTM cell is shown as Fig. 1 (a) and can be formulated as

$$
\begin{aligned}
f_{(t)} &= \sigma\left(W_f \cdot [h_{(t)}, x] + b_f\right) \\
i_{(t)} &= \sigma\left(W_i \cdot [h_{(t)}, x] + b_i\right) \\
\tilde{c}_{(t)} &= \tanh(W_c \cdot [h_{(t-1)}, x] + b_c) \\
c_{(t)} &= f_{(t)} \odot c_{(t-1)} + i^{(t)} \odot \tilde{c}_{(t)} \\
o_{(t)} &= \sigma\left(W_o \cdot [h_{(t-1)}, x] + b_o\right) \\
h^{(t)} &= o^{(t)} \odot \tanh\left(c^{(t)}\right)
\end{aligned}
\tag{1}
$$

where $f_{(t)}$, $i_{(t)}$, $o_{(t)}$ denote the values of forget gate, input gate and output gate at time stamp t, $c_{(t)}$, $h_{(t)}$ denote the cell state and hidden state respectively, σ=1/(1+e^{-x}), W_* means the weight matrices, b_* denotes the biases, $\tilde{c}_{(t)}$ denotes a candidate value vector and $\odot$ is the element-wise product of the vectors.

The 'S' in the SLSTM means 'sparse-based'. The sparse representation mainly uses the learned sparse coding and its corresponding basis vector to map the feature set from the feature space to the sample data space to complete the expression of the feature on the sample data. To get sparse codes, the optimization problem of the L1 norm needs to be solved first. The optimization problem is formulated as follows

$$
\min_{s} \sum_{i} \| x_i - Bs_i \|_F^2 + \lambda \| s_i \|_1 \tag{2}
$$

where $B = [b_1, b_2, \cdots, b_{d_s}] \in \mathbb{R}^{d_x \times d_s}$ is a dictionary matrix composed of base vectors, $X = [x_1, x_2, \cdots, x_n] \in \mathbb{R}^{d_x \times n}$ is a data matrix, $S = [s_1, s_2, \cdots, s_n] \in \mathbb{R}^{d_s \times n}$ is the sparse codes matrix to be generated, and λ is the proportional coefficient.

ISTA is a method widely used to solve (2). In ISTA, the optimization problem of sparse coding is divided into two parts, $\sum_i \|x_i - Bs_i\|_F^2$ and $\|s_i\|_1$. The former part, which is the reconstruction error term, is optimized by the gradient descent method. The latter part, that is, the sparse penalty term is optimized by the hard-thresholding operator. The overall optimization function can be expressed as

$$
s^{(t)} = sh_{(\lambda\tau)}\left(s^{(t-1)} - \tau \nabla g\left(s^{(t-1)}\right)\right) \tag{3}
$$

where $sh_{(\lambda\tau)}(s) = sign(s)\left(|s| - \lambda\tau\right)_+$, According to the update rules, (3) turns to

$$
s^{(t)} = sh_{(\lambda\tau)}\left(s^{(t-1)} - \tau\left(B^T\left(Bs^{(t-1)} - X\right)\right)\right) \tag{4}
$$

when $W_e = I - \tau B^T B$, $W_d = \tau B^T$, then (4) turns to

$$
s^{(t)} = sh_{(\lambda\tau)}\left(W_e s^{(t-1)} + W_d X\right) \tag{5}
$$

In order to fully consider historical information and improve the adaptability of the model, adaptive momentum vectors $i^{(t)}$ and $f^{(t)}$ at time stamp t was introduced, (5) can be reformulated as

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$$\begin{aligned}\tilde{c}^{(t)} &= W_e s^{(t-1)} + W_d x \\ c^{(t)} &= f^{(t)} \odot c^{(t-1)} + i^{(t)} \odot \tilde{c}^{(t)} \\ s^{(t)} &= sh_{(\lambda\tau)}(c^{(t)})\end{aligned} \tag{6}$$

Note that (6) is highly correlated with (1) as $i^{(t)}$ and $f^{(t)}$ corresponding to input gate and forget gate in LSTM, respectively. Recasting (6) according to the structure of LSTM and then there comes the unit of SLSTM.

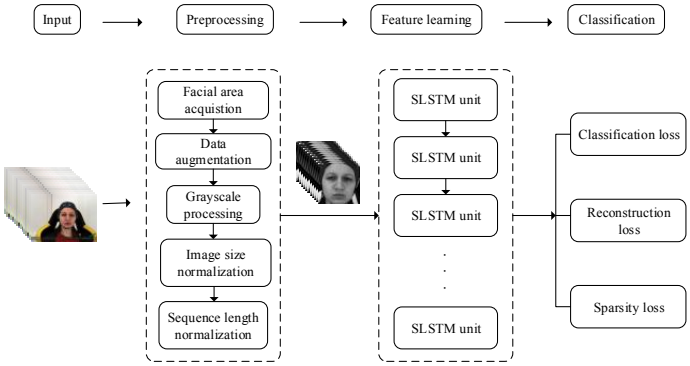


Fig. 2. The overall process of SLSTM for facial expression recognition.

B. How does SLSTM work

To deal with the vanishing gradient problem, the shrink function is replaced with the Double Tanh function $h_{(D,u)} = D(\tanh(x+u) + \tanh(x-u))$. The unit of SLSTM is shown as Fig. 1 (b) and can be formulated as

$$\begin{aligned}i^{(t)} &= \sigma(W_{is} s^{(t-1)} + W_{ix} x) \\ f^{(t)} &= \sigma(W_{fs} s^{(t-1)} + W_{fx} x) \\ \tilde{c}^{(t)} &= W_e s^{(t-1)} + W_d x \\ c^{(t)} &= f^{(t)} \odot c^{(t-1)} + i^{(t)} \odot \tilde{c}^{(t)} \\ s^{(t)} &= h_{(D,u)}(c^{(t)})\end{aligned} \tag{7}$$

u denotes a trainable vector and D denotes a diagonal matrix. $c^{(t)}$ retains all the important information of the network so far, and determines the sparse code to be output at this moment through the Double Tanh function.

In (7), we can see that $\tilde{c}^{(t)} = W_e s^{(t-1)} + W_d x$ and $s^{(t)} = h_{(D,u)}(c^{(t)})$ are the core of ISTA, which guarantees the sparsity of feature representation, realizes the automatic selection of features, and brings interpretability to SLSTM. $c^{(t)} = f^{(t)} \odot c^{(t-1)} + i^{(t)} \odot \tilde{c}^{(t)}$ is the core of LSTM, which utilizes historical information and also enables automatic filtering of information.

C. SLSTM for facial expression recognition

The final output sparse code is multiplied by the learned classification matrix $C \in \mathbb{R}^{d_s \times c}$, and the softmax $(\cdot)$ function is used for multi-class recognition

$$\text{soft}\max(Cs) \tag{8}$$

The final loss function is obtained by combining reconstruction error, sparse error and classification error.

The overall facial expression recognition flow chart is shown in Fig. 2 (no facial area acquisition in CASME II and SMIC).

IV. Experiments and Analysis

A. Databases

This article introduced SLSTM into facial expression recognition and verified its effectiveness for different types of expressions, including pain and micro-expressions. The databases used in our experiments included the Biovid Heat Pain Database (Part A) [15], the CASME II database [16] and the SMIC database [17].

The Biovid Heat Pain Database (Part A) contains 8,700 videos from 87 subjects and the pain rating is divided into five categories. BL1 indicates baseline, PA1 means pain threshold (change from heat to pain), PA4 is the tolerance threshold (pain becomes unacceptable), PA2 and PA3 are two intermediate pain levels.

The CASME II database includes 247 spontaneous facial micro-expressions from 26 subjects. There are five classes of the micro-expressions in this database: happiness (32 samples), surprise (25 samples), disgust (64 samples), repression (27 samples) and others (99 samples).

The SMIC database contains three subsets: SMIC-HS (of all 16 subjects), SMIC-VIS (of the last 8 subjects), and SMIC-NIR (of the last 8 subjects). The SMIC-HS consists of 164 spontaneous micro-expressions, and these frame sequences are categorized into positive (51 samples), negative (70 samples) and surprise (43 samples) classes. Both SMIC-VIS and SMIC-NIR consist of 71 spontaneous micro-expressions, which are divided into positive (28 samples), negative (23 samples) and surprise (20 samples) categories.

B. Data Preprocessing

A series of preprocessing were conducted on the three databases, mainly included the facial area acquisition, data augmentation, grayscale processing, sequence length normalization and Image size normalization. In the Biovid database, each sample video has a duration of 5.5s and a total of 138 frames. In order to fully explore the data, we used the six as the interval and took the 1, 2, 3, 4, 5, and 6 frames as the starting frames respectively to intercept the sample video. In addition, the videos in the Biovid were not pure expression area, we called the Haar classifier in OpenCV for face detection and then intercepted the facial image.

For CASME II and SMIC databases, the steps of facial area acquisition were eliminated since the samples in both datasets were purely facial. However, each facial sample was rotated by -7°, -5°, -3°, 3°, 5°, 7°, and the total number of samples was expanded by a factor of seven, so that to solve the sample limitation problem

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In addition, considering that the color information had little positive effect on facial expression recognition, and even affected the robustness of the algorithm and improved the computational complexity, we made the images in these three databases grayscale. We also normalized the size of these images, and the processed image size is 32×32. Moreover, as each sample in CASME II and SMIC was of varying lengths, the nearest neighbor interpolation method was used to normalize the sequence length to 10 on three databases. The selection of the sequence length was referenced to [18].

C. Pain Recognition Based on Facial Expression

The database used in this section was the Biovid (Part A). The main content was SLSTM model parameter optimization and human-independent experiments with leave-one-group-out (LOGO) cross validation.

We randomly selected 75% of the samples in each category of each subject as the train set, and the remaining 25% were used as the test set. Due to the adaptability of SLSTM, only the number of layers and the size of the batch needed manual optimization. Table 1 showed the experimental results and 'All vs All' means five-category classification. Fig. 3 showed the confusion matrix for the five-category task, 0 corresponded to BL1 and 1-4 respectively corresponded to PA1-PA4.

TABLE I. THE RESULTS (%) OF DIFFERENT NETWORK PARAMETERS

Considered classes	**All vs All**	
Layers	***Batch(128)***	***Batch(256)***
1	**32.0**	28.9
2	24.2	28.5
3	25.8	28.1
Chance	20.0	

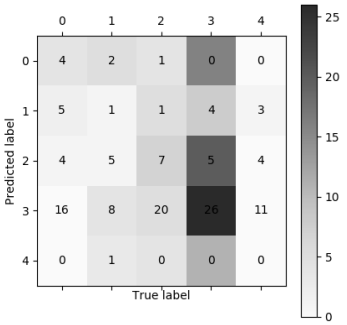


Fig. 3. The confusion matrix for five-category classification on Biovid.

It can be seen that when batch size was 128 and the number of layers was 1, the network achieved the best performance. These parameters were used in all subsequent experiments.

Considering the actual application scenario, the facial expression recognition in this paper was mainly for the person-independent case. As there were so many subjects in Biovid, we used a more rigorous method, which named leave-one-group-out (LOGO) cross-validation. 87 subjects was randomly divided into 8 groups (7 groups of 11 people, 1 group of 10 people), then one group was randomly picked as test set and the others as train set. The results of pain detection (pain/no pain) between four pain levels and baseline and "All vs All" recognition on the Biovid database were shown as Table 2.

TABLE II. THE RECOGNITION RESULTS ON THE BIOVID DATABASE

Method	Considered classes				
	PA1 vs BL1	PA2 vs BL1	PA3 vs BL1	PA4 vs BL1	All vs All
Chance	50.0	50.0	50.0	50.0	20.0
BoTF-SVM[19]	49.5	51.1	60.7	71.2	26.3
FAD-RF[20]	53.3	56.0	**64.0**	**72.4**	**30.8**
SLSTM	**56.5**	**57.1**	59.6	61.7	29.7

The data partitioning methods in the comparison results were leave-one-subject-out (LOSO) cross validation. M Bartlett et al. [19] recently proposed an approach with a Bag of Temporal Features (BoTF) descriptor, and an SVM with RBF kernel was used for classification. In [20], Werner et al. proposed a descriptor named Facial Activity Descriptor (FAD), and the Random Forest (RF) classifier was used for the classification.

It can be seen that although we used a more stringent data partitioning method than these references, SLSTM still obtained a not bad performance in the five-category task and even 3.4% higher than [19], which proposed a traditional hand-crafted method. Compared to other methods, SLSTM exceled in low-intensity pain detection and increased accuracy by 3.2% and 1.7% respectively, which reflected the superiority of SLSTM in describing subtle facial expressions. Although the effect was not satisfactory in the detection of high-intensity pain, it also showed applicability in the pain recognition. In addition, compared to the LOSO cross validation, the LOGO cross validation has simpler train sets and more complex test sets, which also brings room for the improvement of the recognition accuracy of SLSTM.

D. Facial Micro-expression Recognition

Based on the LOSO cross validation, we performed facial micro-expression categories recognition experiments on CASME II and SMIC, and compared the results of SLSTM with the results of LSTM with an equivalent structure.

TABLE III. THE RECOGNITION ACCURACY (%) OF SLSTM ON FACIAL EXPRESSION RECOGNITION

Method	**SMIC**			**CASME II**
	HS	***NIR***	***VIS***	
Chance	33.3	33.3	33.3	20.00
LBP-SIP[21]	42.7	-	-	53.9
LOCP-TOP[22]	45.7	-	-	**56.7**
OSW[23]	53.1	-	-	41.7
FDM[24]	54.9	-	-	42.0
SLSTM	**57.8**	**46.3**	**54.0**	54.6

The results of facial micro-expression categories recognition experiments on CASME II and SMIC were shown in Table 3. The comparative experimental results were all based on LOSO cross validation, using Local Binary Pattern-Three Orthogonal Planes (LBP-SIP) [21], Local Ordinal Contrast Pattern-Three Orthogonal Plane (LOCP-TOP) [22], Optical Strain Weighted (OSW) [23], Facial Dynamics Map (FDM) [24] for feature extraction respectively, and classified by the Convolutional Neural Network as described in [11]. SLSTM achieved 2.9% better performance than the best available results [24] on SMIC-HS and also achieved acceptable results (54.0%) on SMIC-VIS. The reason for the poor performance on SMIC-NIR may be the fact that this subset was obtained by a near infrared camera, which may

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interfere with the feature extraction to some extent. The performance on CASME II was also superior to most of other methods, only 2.1% worse than LOCP-TOP.

From Table 4, we could see that SLSTM had a great advantage in recognition accuracy compared to LSTM with corresponding structure, which reflected the advantages of sparsity in the feature learning of facial micro-expression clearly.

TABLE IV. COMPARISON OF SLSTM AND LSTM

Method	SMIC-HS	CASME II
Chance	33.3	20.0
LSTM	29.9	39.9
SLSTM	**57.8**	**54.6**

V. EXPERIMENTS AND ANALYSIS

In this paper, the sparse representation was introduced into facial feature learning and dynamic facial expression recognition, and the SLSTM with sparsity and interpretability based on LSTM was obtained. The experimental results on Biovid, CASME II and SMIC demonstrated the superiority of this method. In the future, considering the complexity of the information of facial expression sequences, CNN will be used to extract the features first and then input these features into SLSTM to improve the recognition accuracy. In addition, the relationship between ISTA and other RNN variants will be further explored to obtain a more efficient network structure in anticipation of the two-way improvement of interpretability and accuracy.

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